

## Lab 4

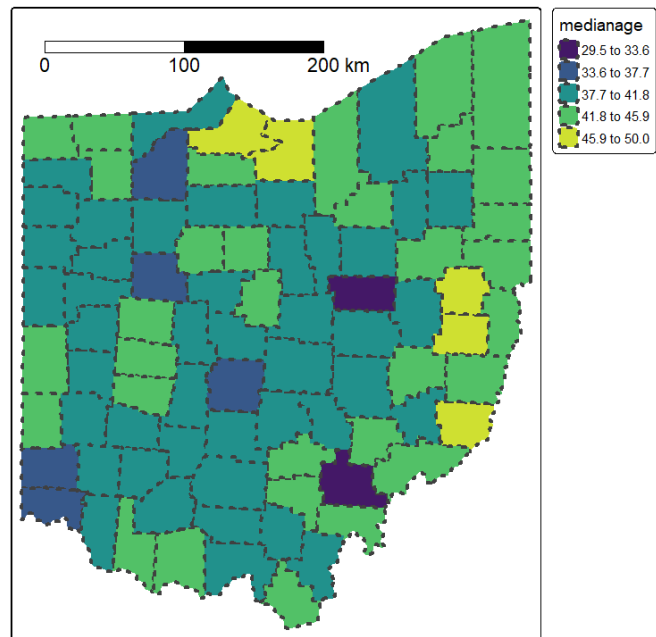
### Environmental Analysis in R

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#### QA-1: My choices in making map 1

**Ohio Scale:** *Joining spatial and tabular data, making a choropleth map using a variable of my interest, customizing border, adding a scale-bar*

To complete this task, I joined `oh_counties.gpkg` (spatial data) with `oh_counties_DP2020` (tabular data). Before joining, I removed one row from the table representing each variable's total for Ohio. Then, I joined them using `GEOIDFQ` and `geoid` as the primary key field. For making the choropleth map, I used the variable “medianage”, and to customize the border, I changed the line style and width. Finally, I added the scale bar.

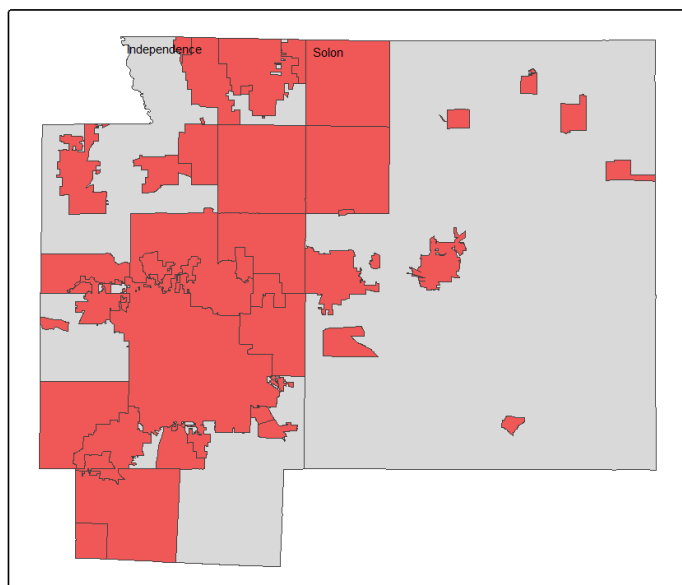


#### **Local Scale:**

**Part 1:** *Creating municipal boundaries within Portage and Summit Counties, using names for labeling*

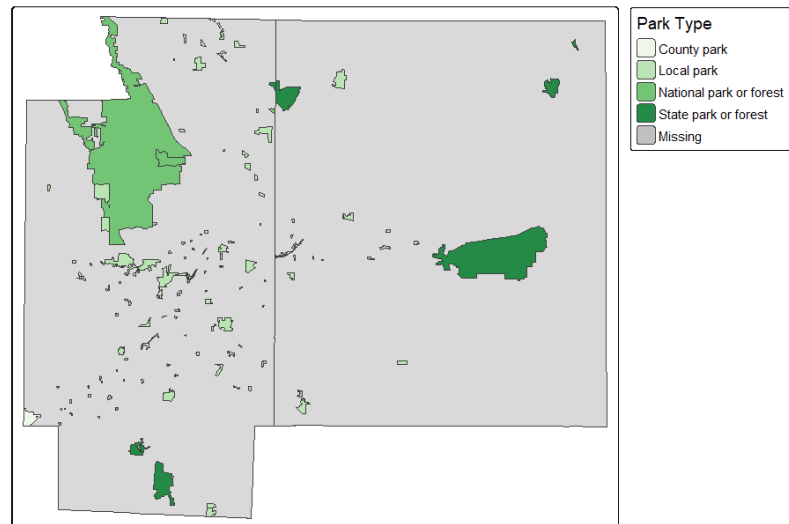
This part required some data organization, such as extracting the Portage and Summit Counties from the Ohio counties. Then, after checking the projection of two different spatial data files, I identified the municipal areas within the selected counties, using the datafile `oh_places.gpkg`. For labeling, I used the “Name” variable.

\*\*\*I tried multiple times using different codes, but for some reason, all the labels didn't show up.



**Part 2: Parks within Portage AND Summit counties, symbolized using different shades of green according to the park TYPE**

This part also required some data organization, such as matching the projection of the park dataset (oh\_parks.gpkg) with the selected county dataset. Then, I extracted the parks within the selected counties and differentiated them with shades of green, based on the variable “FEATTYPE”, which is the type of park.

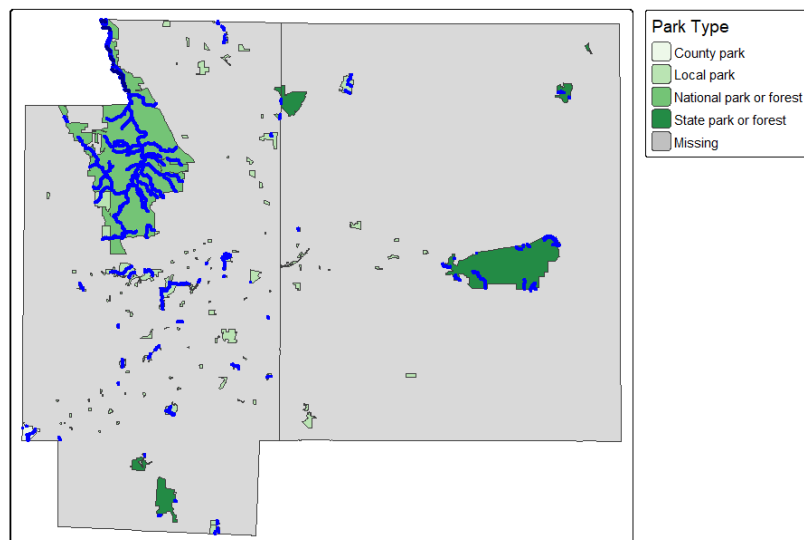


**Part 3: Linear water features (streams, rivers) in Portage AND Summit counties. Symbols indicating which linear features intersect a park**

In this part, I used both the river dataset (oh\_rivers.gpkg) and linear water feature datasets (tl\_2022\_39133\_linearwater.shp & tl\_2023\_39153\_linearwater.shp), although finally, I identified that tl\_2023\_39153\_linearwater had the river in it that I got from the oh\_rivers.gpkg.

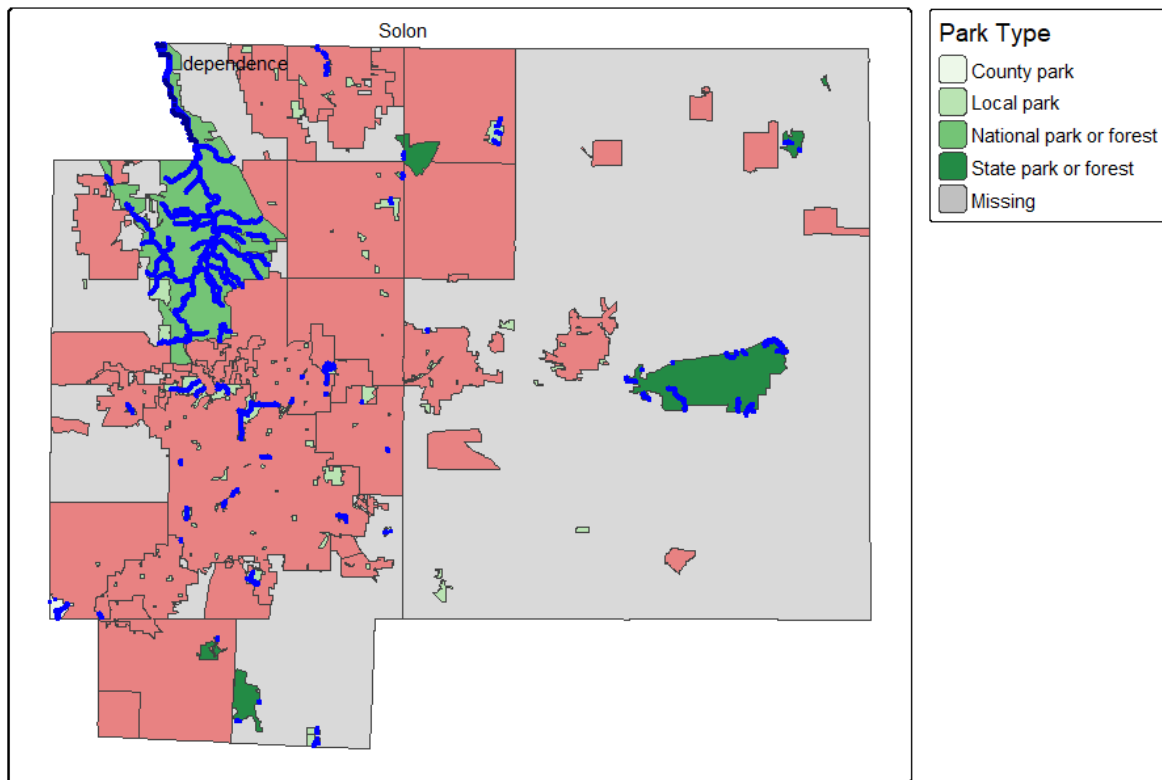
For this task, I projected all the data files into one projection and exported the streams and rivers that intersected the parks I used in part 2.

\*\*\*Here, Blue represents the intersected linear streams, and dark blue (in the top left corner) represents the intersected river with parks.



**Putting all of them together (Part 1 to 3)**

Finally, I put all the maps I made from parts 1 to 3 on top of each other. First, I added selected county boundaries, then added municipal areas, park maps, and intersected linear water features, including both streams and rivers.



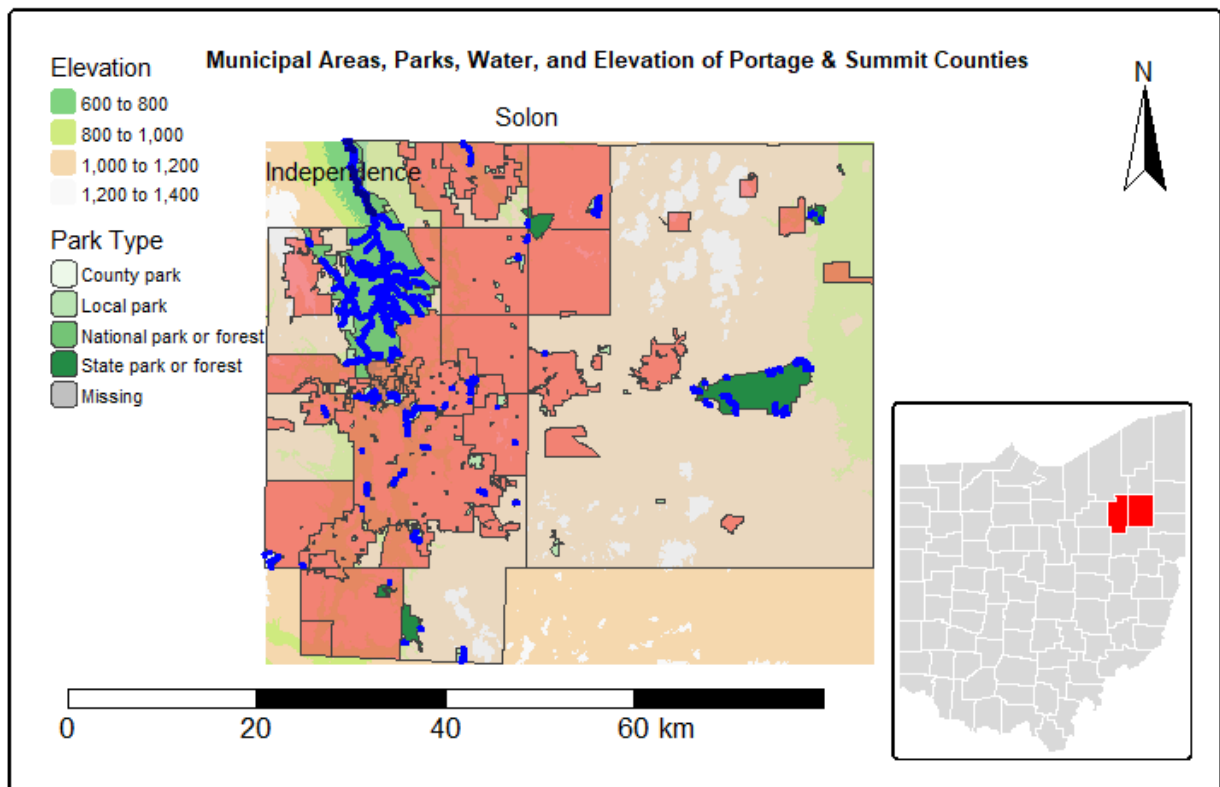
**Final Part:** Plot elevation using DEM data behind a semi-transparent Portage and Summit counties, adding a north arrow, making the Ohio-scale map an inset into the map detailing the Portage + Summit counties study area, adding a title

First, I made a transparent selected county data file, then organized the DEM data file, which included the DEM data projection to the desired coordinate system and crop, using the extent of selected county file. Then I created the map of dem data and added the title and north arrow. Finally, I put together all the layers (elevation map, transparent county boundary map, and the maps I created in that last task).

\*\*\*I tried to fit the map within the map frame, the best I could do.

\*\*\*When I try to run the whole code all at once, I face the following error just before plotting the final map. And so, I need to run the code of final map plotting again.

```
Error in save(plot, file = filename) : reached elapsed time limit
Graphics error: Plot rendering error
```



## **QA-2:** *Comparing the cartographic decisions*

### **Ohio Scale (Group 1)**

I was in group 1, which was responsible for making the Ohio scale map. Though I didn't find this map very useful in creating the final frankenmap, it helped me to learn how to create a choropleth map and handle different types of datasets. From the perspective of cartographic design, we learnt how to add a scale bar with a map and customize symbology.

### **Local Scale (Group 2)**

In local map processing, group 2 created a map putting different layers all together, where they had to decide the projection system and work on the final map of a data frame handling various map layers. I did the same thing in my process, but I found that they missed two data layers, municipal areas, and the linear water feature, which was in a different folder, while producing the final map. They only used the Ohio River geo package when I looked at their map in class. So, I incorporated all the layers in my map, which I found necessary.

### **Final Map with all components (Group 3)**

From a cartographic perspective, they added the north arrow, title, and locator or inset map and did the final organization. Their focus was on fitting the map and inset map within the map frame. In my process, I made some changes, which included organizing and positioning different elements according to my preference and the use of `tm_layout` to fit the map within the data frame, as I found the viewport hard to handle for main map positioning. But for the

inset map, I utilized the viewport through trial and error. I also made my version of the Ohio locator map that highlighted the counties of the final map within it.

### **Q4-3:** *My choices in making map 2*

I was asked to create a map using both vector and raster data, including spatial data relating to a social process (e.g., political boundaries) and spatial data relating to an environmental process (e.g., water resources).

To accomplish the task, I used the following data-

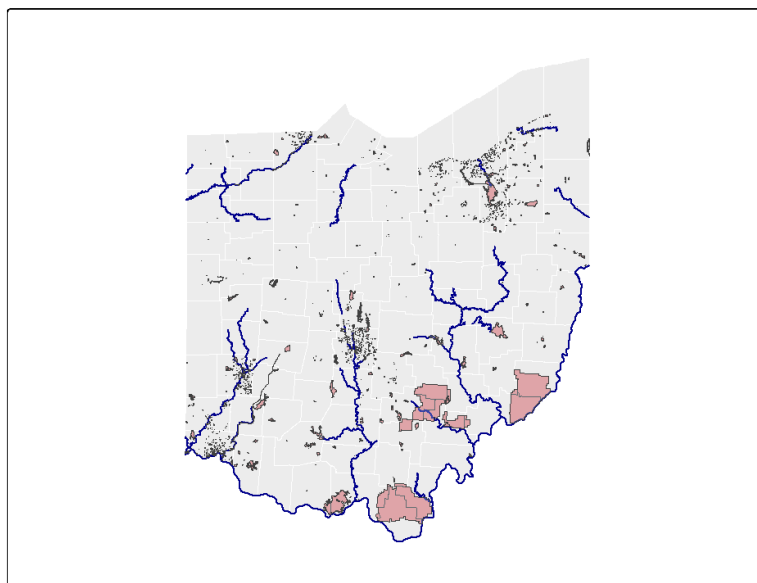
- Ohio counties boundary
- Rivers
- Population data
- DEM data for the whole of Ohio that I downloaded from USGS

Using these datasets, I tried to create a flood risk zone or county map associated with rivers, including the population distribution of those areas, which might require special attention during the implementation of any project or planning in those areas. The steps I followed-

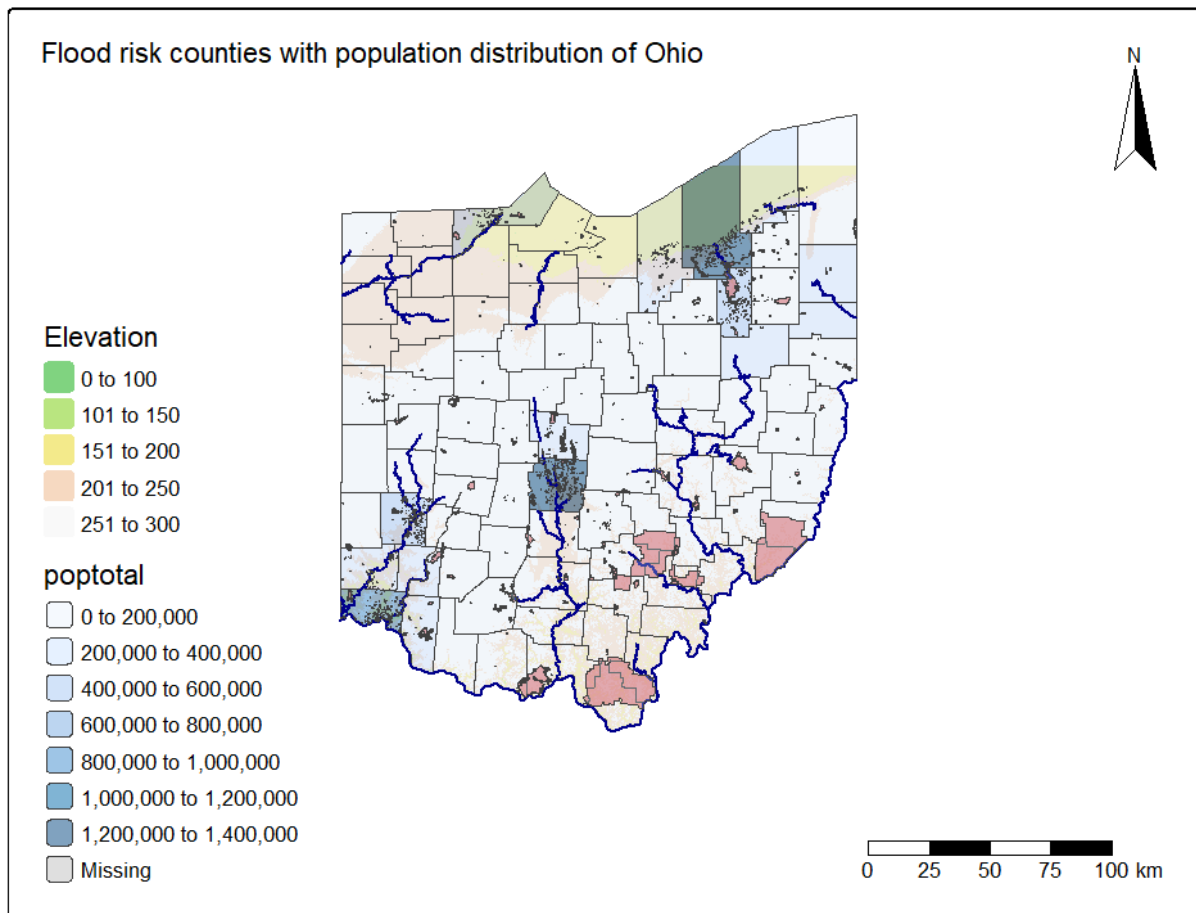
**Step 1:** I made a choropleth of Ohio counties using the variable “poptotal” to understand the population distribution

#### **Step 2:**

- I created a buffer of 1000m from rivers to identify the flood risk areas
- I organized the DEM data, including projection, cropping, and masking, and created a map of areas with low elevation (<50)
- Then, I created an intersected map of the river buffer and low-elevated areas to identify the flood risk zones and later intersect the produced map with the Ohio counties map to find out the counties containing the high flood risk zones, symbolizing them with red color.



**Step 3:** Finally, I put the maps all together with additional map components, such as a north arrow, title, scale, and legend to produce the final map



\*\*\*Here, I didn't consider Lake Erie, which is in the upper right corner, and the counties with transparent red are representing the counties with high flood risk zones.

\*\*\*It takes too much time to run the code and plot the map.

#### QA-4

The package tmap is very useful in map-making because it provides a lot of flexibility to customize the map and its features. Also, it helps to organize the map in a data frame with other map components for visual representation. However, it takes a lot of time to produce a map with multiple layers or with a little bit of complexity. Although I encountered many issues while making my maps that still need to be resolved, I found it very effective. I believe there is a lot to learn about tmap apart from the codes and functions that I have already used.